In [1]:

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# Backpropagation Network for Logic Gates
# Name: Harshal Veer
# Roll No: 75

import numpy as np

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

 def sigmoid_derivative(x):
    return x * (1 - x)

# Input dataset
X_base = np.array([[0,  0],
                    [0,  1],
                    [1,  0],
                    [1,
                    1]])
# Logic gate outputs
 logic_gates = {
    "AND":   np.array([[0],  [0],  [0],  [1]]),
    "OR":    np.array([[0],  [1],  [1],  [1]]),
    "XOR":   np.array([[0],  [1],  [1],  [0]]),
     "NAND": np.array([[1],  [1],  [1],  [0]]),
    "NOR":   np.array([[1],  [0],  [0],  [0]])
}

#Expand dataset
X = np.tile(X_base, (20, 1))

epochs = 5000
learning_rate = 0.5

for gate, Y_base in logic_gates.items():
    Y = np.tile(Y_base, (20, 1))

    # Initialize weights and bias
    np.random.seed(1)
    W1  =  np.random.rand(2,  4)
    W2  =  np.random.rand(4,  1)
    b1 = np.random.rand(4)
    b2 = np.random.rand(1)

    # Training
    for _ in range(epochs):
        hidden_output = sigmoid(np.dot(X, W1) + b1)
        output = sigmoid(np.dot(hidden_output, W2) + b2)

        error = Y - output

        d_output = error * sigmoid_derivative(output)
        d_hidden=  d_output.dot(W2.T) * sigmoid_derivative(hidden_output)

        W2 += hidden_output.T.dot(d_output) * learning_rate
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        W1 += X.T.dot(d_hidden) * learning_rate
        b2 += np.sum(d_output, axis=0) * learning_rate
        b1 += npsum(d_hidden,axis 0)   * learning_rate

    # Testing
    test_output = sigmoid(np.dot(sigmoid(np.dot(X_base, W1) + b1), W2) + b2)

    print("\nLogic Gate:", gate)
    print("Name: Harshal Veer   | Roll No: 75")
    print("A B Output")
    for i in range(4):
        print(X_base[i][0], " ", X_base[i][1], " ", round(test_output[i][0]))
```

```
Logic Gate: AND
Name: Harshal Veer| Roll No: 75
A  B  Output
0   0   0
0   1   0
1   0   0
1   1   1


Logic Gate: OR
Name: Harshal Veer| Roll No: 75
A  B  Output
0   0   0
0   1   1
1   0   1
1   1   1


Logic Gate: XOR
Name: Harshal Veer| Roll No: 75
A  B  Output
0   0   0
0   1   1
1   0   1
1   1   0


Logic Gate: NAND
Name: Harshal Veer| Roll No: 75
A  B  Output
0   0   1
0   1   1
1   0   1
1   1   0


Logic Gate: NOR
Name: Harshal Veer| Roll No: 75
A  B  Output
0   0   1
0   1   0
1   0   0
1   1   0
```

In [ ]: